

Introduction

Yoga is an ancient discipline that integrates the mind, body, and breath to promote physical health, mental well-being, and overall balance. Achieving correct alignment and posture in each yoga pose is essential to fully benefit from the practice and to minimize the risk of injury. However, for beginners and even intermediate practitioners, maintaining accurate posture without external guidance can be challenging.

To address this issue, the Yoga Position Assistant is developed as an intelligent digital platform that supports users in learning, practicing, and refining yoga poses. The application utilizes advanced computer vision techniques, particularly the MediaPipe Pose framework, to detect and track human body landmarks from images and video streams of yoga poses. By comparing the user's posture with predefined reference poses, the system provides real-time visual feedback, highlighting misalignments and suggesting corrective improvements.

This project focuses on building a user-friendly application with a responsive graphical user interface (GUI) implemented using PyQt5. The system allows users to select specific yoga poses, visualize extracted skeletal landmarks. It is designed to support users across different skill levels, from beginners to advanced practitioners, enabling self-guided practice while enhancing safety, accuracy, and effectiveness.